

Chapter 5 Test: Exponential and Trigonometric Functions

Full Name: [REDACTED]	Date: December 1, 2023
Full solutions must be shown for full marks. Solutions must demonstrate methods shown in class.	TOTAL: 29.5/32

Knowledge:

1. The derivative of $f(x) = e^{3x}$ is:

- A) e^{3x} B) e^{2x} C) $3e^{3x}$ D) $3e^{2x}$

2. The derivative of $f(x) = 2^{\sin x}$ is:

- A) $(\cos x)(2^{\sin x})$ B) $(\sin x)(2^{\sin x})$ C) $(\cos x)(\ln 2)(2^{\sin x})$ D) $(-\cos x)(\ln 2)(2^{\sin x})$

3. The derivative of $f(x) = \sin(x^3 - x)$ is:

- A) $(3x^2 - 1)\cos(x^3 - x)$ B) $-(3x^2 - 1)\cos(x^3 - x)$
C) $\cos(x^3 - x)$ D) $-\cos(x^3 - x)$

4. The function $f'(x) = \frac{1}{2x}$ is the derivative function of:

- A) $\ln \sqrt{x}$ B) $\sqrt{\ln x}$ C) $\ln x^2$ D) $\ln x$

5. The derivative of $f(x) = \log(x^3)$ is:

- A) $\frac{3}{\ln 10}$ B) $\frac{3x^2}{\ln 10}$ C) $\frac{2}{x \ln 10}$ D) $\frac{3}{x \ln 10}$

6. The derivative of $f(x) = \cot x$ is:

- A) $-\frac{1}{(\cos^2 x)}$ B) $-\frac{1}{\sin^2 x}$ C) $\tan^2 x$ D) $\frac{1}{\sin^2 x}$

7. The derivative of $f(x) = x \sin x$ is:

- A) $1 + \cos x$ B) $1 - \cos x$ C) $\sin x + x \cos x$ D) $\sin x - \cos x$

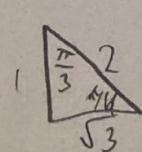
8. The equation $0 = e^{-x^2}(2\cos 2x - \sqrt{3})$ on the interval $[0, 2\pi]$ has solutions:

- A) $x = 0, \pi, 2\pi$ B) $x = \frac{\pi}{6}, \frac{11\pi}{6}$ C) $x = \frac{\pi}{12}, \frac{11\pi}{12}$ D) No Solutions

$$\cos 2x = \frac{\sqrt{3}}{2}$$

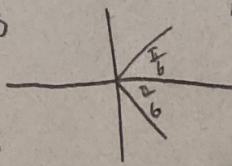
$$\cos a = \frac{\sqrt{3}}{2}$$

$$a = \frac{\pi}{6}, \frac{11\pi}{6}$$



$$2x = \frac{\pi}{6}$$

$$x = \frac{\pi}{12}, \frac{11\pi}{12}$$



8

9. Determine the equation for the tangent line to the curve $y = 5^{-x^2}$ when $x=1$. Use exact answers.

$$x=1 \quad y=5^{-1} = \frac{1}{5}$$

$$\frac{dy}{dx} = -2x(\ln 5)(5^{-x^2})$$

$$\left. \frac{dy}{dx} \right|_{x=1} = -2(1)(\ln 5)(5^{-1}) = -\frac{2\ln 5}{5}$$

$$\therefore y = -\frac{2\ln 5}{5}x + \frac{2\ln 5 + 1}{5}$$

$$y = -\frac{2\ln 5}{5}x + b \quad \rightarrow \quad 1 = -2\ln 5 + 5b$$

$$\frac{1}{5} = -\frac{2\ln 5}{5}(1) + b \quad \rightarrow \quad 5b = 1 + 2\ln 5$$

$$b = \frac{1 + 2\ln 5}{5}$$

Application:

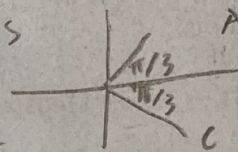
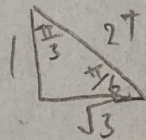
10. Find the global extrema for $f(x) = \frac{x}{2} - \sin x$ on the interval $[0, 2\pi]$

$$f'(x) = \frac{1}{2} - \cos x$$

$$0 = \frac{1}{2} - \cos x$$

$$\cos x = \frac{1}{2}$$

$$x = \frac{\pi}{3}, \frac{5\pi}{3}$$



$$f\left(\frac{5\pi}{3}\right) = \frac{5\pi}{6} - \sin\left(\frac{5\pi}{6}\right)$$

$$= \frac{5\pi}{6} + \frac{\sqrt{3}}{2} = 3.48$$

$\therefore$ Abs Max at $\left(\frac{5\pi}{3}, 3.48\right)$
Abs Min at $\left(\frac{\pi}{3}, -0.34\right)$

$$f\left(\frac{\pi}{3}\right) = \frac{\pi}{6} - \sin\left(\frac{\pi}{3}\right) = \frac{\pi}{6} - \frac{\sqrt{3}}{2} = -0.34$$

$f(0) \quad f(2\pi)$

11. Find the parameter a such that the function $y = \sin(ax)$ satisfies the relation: $y + 3y'' = 0$

$$y' = a \cos(ax)$$

$$y'' = -a^2 \sin(ax)$$

$$y + 3y'' = 0$$

$$\sin(ax) + 3a^2 \sin(ax) = 0$$

$$\sin(ax) = -3a^2 \sin(ax)$$

$$1 = -3a^2$$

$$a^2 = -\frac{1}{3}$$

$$a = \pm \frac{1}{\sqrt{3}} = \pm \frac{\sqrt{3}}{3}$$

$$\therefore a = \frac{\sqrt{3}}{3} \text{ or } a = -\frac{\sqrt{3}}{3}$$

$a = 0$!

8.5

12. A new product has just come onto the market and is a big hit. The success, however, does not last too long. Within a year, the number of sales per day has dropped drastically. Given that the number of sales per day, measured in tens of thousands, of the product is represented by the function $n(t) = -50(e^{-4t} - e^{-3t})$ after t years, how many products were sold on the busiest day of the year?

$$n'(t) = -50(-4e^{-4t} + 3e^{-3t})$$

$$= -50e^{-4t}(-4 + 3e^t)$$

$$= -50e^{-4t}(3e^t - 4)$$

none

$$3e^t = 4$$

$$e^t = \frac{4}{3}$$

$$t = \ln \frac{4}{3}$$

$$= 0.29$$

$$n\left(\ln \frac{4}{3}\right) = -50\left(e^{-4\ln \frac{4}{3}} - e^{-3\ln \frac{4}{3}}\right)$$

$$= -50\left(\left(\frac{4}{3}\right)^{-4} - \left(\frac{4}{3}\right)^{-3}\right)$$

$$= -50\left(\frac{81}{256} - \frac{27}{64}\right)$$

$$= -50\left(\frac{-27}{256}\right) = \frac{1350}{256}$$

$$= 5.2734$$

$\therefore 52,734$ products were sold on the busiest day of the year

Communication:

13. Explain how you could go about differentiating $f(x) = \csc x$ in two different ways. (Hint: The two methods are very similar).

You can make $\csc x = \frac{1}{\sin x}$ first, then there are 2 methods:-

1. Use quotient rule $\rightarrow \frac{(0)\sin x - (1)\cos x}{\sin^2 x} = \frac{-\cos x}{\sin^2 x} = -\cot x \csc x$

2. Make $\frac{1}{\sin x} = (\sin x)^{-1}$ and use chain rule $\rightarrow -1(\sin x)(\cos x) = \frac{-\cos x}{\sin^2 x} = -\cot x \csc x$

14. Find the derivative of $f(x) = 2^{x^2}$ and $g(x) = e^{x^2}$. Explain what looks different about these derivatives and why this is the case.

$$f'(x) = 2x(\ln 2)(2^{x^2})$$

$$g'(x) = 2x e^{x^2}$$

The obvious difference is that one is base 2 while the other is base e . But the important difference is that because one is base e , it's missing the $\ln e$ while the base 2 has a $\ln 2$. This is because the $\ln e$ is actually there, $\ln e$ is equal to 1, anything multiplied by 1 is itself, therefore there's no need to write it out

Thinking:

15. Determine the critical values (you may use approximate radians) of the function $f(x) = (\sin x)e^{\cos x}$ on the interval $[0, 2\pi]$

$$f'(x) = (\cos x)e^{\cos x} + \sin x(-\sin x)e^{\cos x}$$

$$= \cos x e^{\cos x} - \sin^2 x e^{\cos x}$$

$$0 = e^{\cos x} (\cos x - \sin^2 x)$$

$$\cos x - \sin^2 x = 0$$

$$\cos x - (1 - \cos^2 x) = 0$$

$$\cos x + \cos^2 x - 1 = 0$$

$$a = -1 \pm \frac{\sqrt{1 - 4(1)(-1)}}{2} = \frac{-1 \pm \sqrt{5}}{2} = 0.62, -1.62$$

$$a = 0.62$$

$$\cos x = 0.62$$

$$x = 0.9$$

$$x = 0.62, 5.66$$

$$f(0) = (\sin 0)(e^{\cos 0}) = 0$$

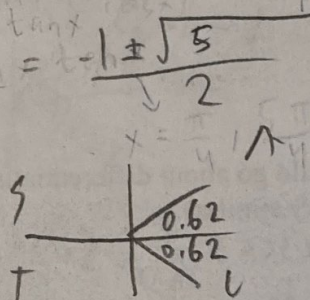
$$f(2\pi) = (\sin 2\pi)(e^{\cos 2\pi}) = 0$$

$$f(0.62) = 1.31$$

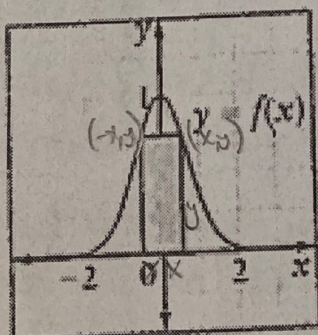
$$f(5.66) = -1.31$$

$$a = \cos x$$

$$a^2 + a - 1$$



16. One side of a rectangle lies on the x-axis. The opposite side of the rectangle has vertices which lie on the graph of $f(x) = e^{-x^2}$. For what value of x is the area of the rectangle a maximum?



$$A = 2xy$$

$$A(x) = 2x(e^{-x^2})$$

$$A'(x) = 2e^{-x^2} + 2x(-2x)(e^{-x^2})$$

$$A'(x) = 2e^{-x^2} - 4x^2(e^{-x^2})$$

$$A'(x) = 2e^{-x^2}(1 - 2x^2)$$

$$A\left(\frac{\sqrt{2}}{2}\right) = \sqrt{2} \left(e^{-\frac{1}{2}}\right) = 0.86$$

$$A\left(-\frac{\sqrt{2}}{2}\right) = -\sqrt{2} \left(e^{-\frac{1}{2}}\right) = -0.86$$

$$\therefore x = \frac{\sqrt{2}}{2}$$

$$1 = 2x^2$$

$$x^2 = \frac{1}{2}$$

$$x = \pm \frac{1}{\sqrt{2}} = \pm \frac{\sqrt{2}}{2}$$

no neg.